

§ 2 Matrix alg.

We have seen that matrices can be quite useful.

One powerful application is for the computer models for aircraft design. It uses a lot of large scale matrix. We will start from the basics.

§ 2.1 Matrix operations

Basic notions

$A: m \times n$ matrix

We write

$$A = [\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n] = \begin{bmatrix} a_{11} & \dots & a_{1j} & \dots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \dots & a_{mj} & \dots & a_{mn} \end{bmatrix} \quad \text{row } i$$

$$\vec{a}_1, \dots, \vec{a}_n \in \mathbb{R}^m$$

$$\vec{a}_j \quad \text{column } j \quad \vec{a}_n$$

a_{ij} is called the (i, j) -entry of A .

diagonal entries of A : $a_{11}, a_{22}, \dots$

They form the main diagonal of A .

A diagonal matrix is a square $n \times n$ matrix whose nondiagonal entries are 0:

$$\begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{bmatrix}$$

An $m \times n$ matrix whose entries are all zero is a zero matrix and is written as O .

Easy operations: sums and scalar multiples

Equal Two matrices are equal if they have the same size and their corresponding entries are equal. i.e. corresponding columns are equal.

Sums If A and B are $m \times n$ matrices, then the sum $A+B$ is the $m \times n$ matrix whose (i,j) -entry is the sum of the two (i,j) -entries of A and B .

Use this $\rightarrow$ $A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{m \times n}$ $A+B = (a_{ij} + b_{ij})_{m \times n}$
i.e. the columns are the sums of the corresponding columns in A and B .

The sum $A+B$ is defined only when A & B are the same size.

Exmp Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Then $A+B = \begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}$, $A+C$ is NOT defined.

Scalar multiple $r \in \mathbb{R}$, $A = (a_{ij})_{m \times n}$ $\swarrow$ (i,j) -entry

$rA := (ra_{ij})_{m \times n}$ definition

Notation $-A = (-1)A$

Exmp A, B as above $2A - B = \begin{bmatrix} 2 \times 1 - 0 & 2 \times 2 - 1 \\ 2 \times 3 - 1 & 2 \times 4 - 2 \end{bmatrix}$
 $= \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix}$

Thm 1 A, B, C $m \times n$ matrices, $r, s \in \mathbb{R}$

b) $A+0 = A$, c) $A+(-A) = 0$, a) $(A+B)+C = A+(B+C)$

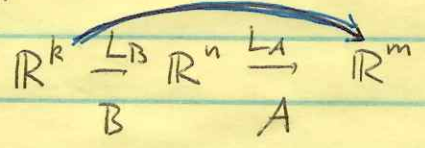
d) $A+B = B+A$, (e) $r(sA) = (rs)A$ (f) $r(A+B) = rA+rB$

(g) $(r+s)A = rA+sA$

Matrix multiplication

omit
this

lin. maps



Standard
matrices

Sizes $n \times k$ $m \times n$

$$\vec{x} \mapsto B\vec{x} \mapsto A(B\vec{x}) = (AB)\vec{x}$$

Some matrix,
which we denote
by AB .

So, the composition (blue arrow) is multiplication by AB .
The AB is also linear!!!

Q. $AB = ?$

AB is the standard matrix of the composition $L_A \circ L_B$.
First, it should be an $m \times k$ matrix, because it corresponds to

a map $\mathbb{R}^k \rightarrow \mathbb{R}^m$.
Denote $B = [b_{11} \dots b_{1k}]$

We study the images of $\vec{e}_1, \dots, \vec{e}_k$ under the composition

$$\vec{e}_1 \mapsto B\vec{e}_1 \mapsto A(B\vec{e}_1) = A\vec{b}_1 \in \mathbb{R}^m$$

$$= \vec{b}_1$$

$\vdots$

$$\vec{e}_k \mapsto B\vec{e}_k \mapsto A\vec{b}_k$$

$$= \vec{b}_k$$

Thus, $AB = [A\vec{b}_1 \dots A\vec{b}_k]$

$A: m \times n$ $B: n \times k$

$m \times k$ matrix



Row-column rule for computing AB

$$A = (a_{ij})_{m \times n} \quad B = (b_{ij})_{n \times k}$$

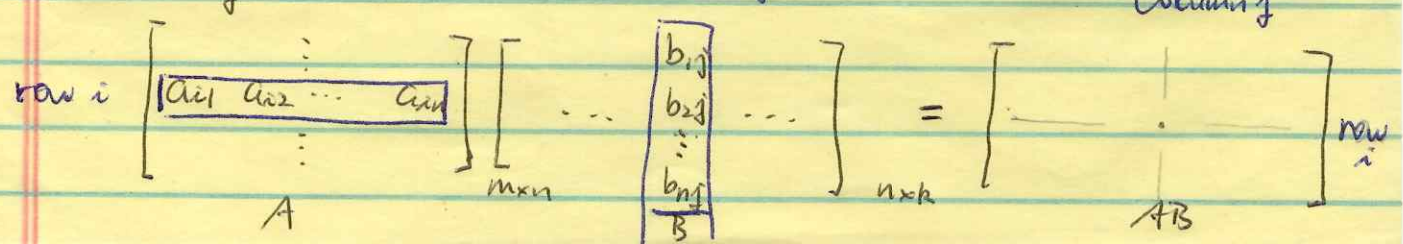
AB is defined like this
so that it corresponds to
the composition of
2 linear maps.

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$$

Pictorially

column j

column j



Exmp $A = \begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}$

$$AB = \begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \end{bmatrix}$$

BA NOT defined.

Properties of Matrix multiplication

Recall. I_m denotes the $m \times m$ identity matrix

Thm 2 A $m \times n$ matrix. B, C s.t. the following are defined

- a) $(AB)C = A(BC)$ (associative)
- b) $A(B+C) = AB+AC$
- c) $(B+C)A = BA+CA$ } (distributive)
- d) $rAB = (rA)B = A(rB) \quad \forall r \in \mathbb{R}$
- e) $I_m A = A = A I_n$

Warning. In general, $AB \neq BA$

Exmp $\begin{bmatrix} * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * \\ * \\ * \end{bmatrix}$ defined $\begin{bmatrix} * \\ * \\ * \end{bmatrix} \begin{bmatrix} * & * & * \\ * & * & * \end{bmatrix}$ not defined.

Exmp $\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$ *

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 4 & 2 \end{bmatrix}$

Warning. Cancellations don't hold in general. That for mult.

$$AB = CB \not\Rightarrow A = C$$

$$\bullet AB = 0 \not\Rightarrow A = 0 \text{ or } B = 0$$

Power $A^k = \underbrace{A \cdot A \cdots A}_k$ $A^0 = I$ identity matrix.
 A square matrix k

Transpose $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$ $A^T = \begin{bmatrix} a_{11} & \cdots & a_{m1} \\ a_{12} & & \vdots \\ \vdots & & a_{mn} \end{bmatrix}$ transpose

Exmp $A = \begin{bmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{bmatrix}$ $A^T = \begin{bmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{bmatrix}$

For examples in computer science or AI. see Exmp 9, 10.